

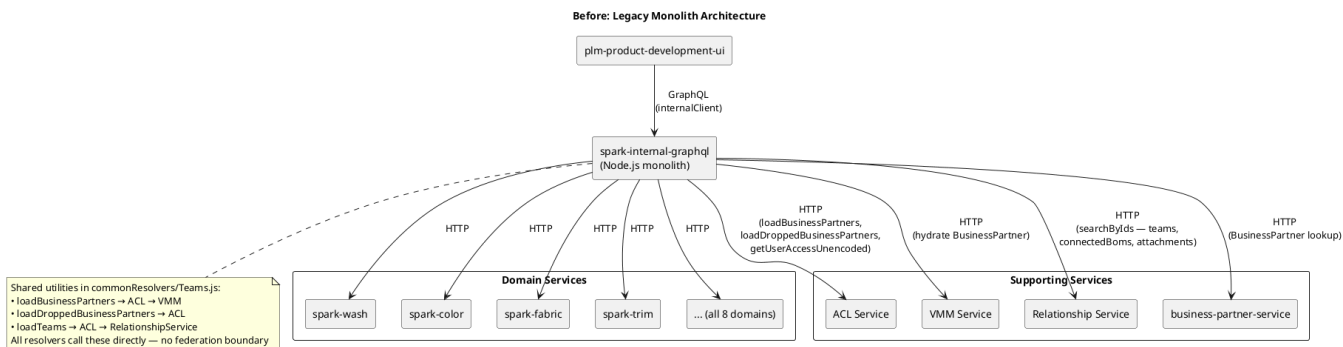
PLM Federation Architecture — Context Reference

This document provides architectural context for ADR review discussions. It is referenced by:

- **ADR-010** — Relationship Service association resolution (teams, connectedBoms, attachments)
- **ADR-011** — BusinessPartner / droppedPartnerIds resolution via ACL and VMM

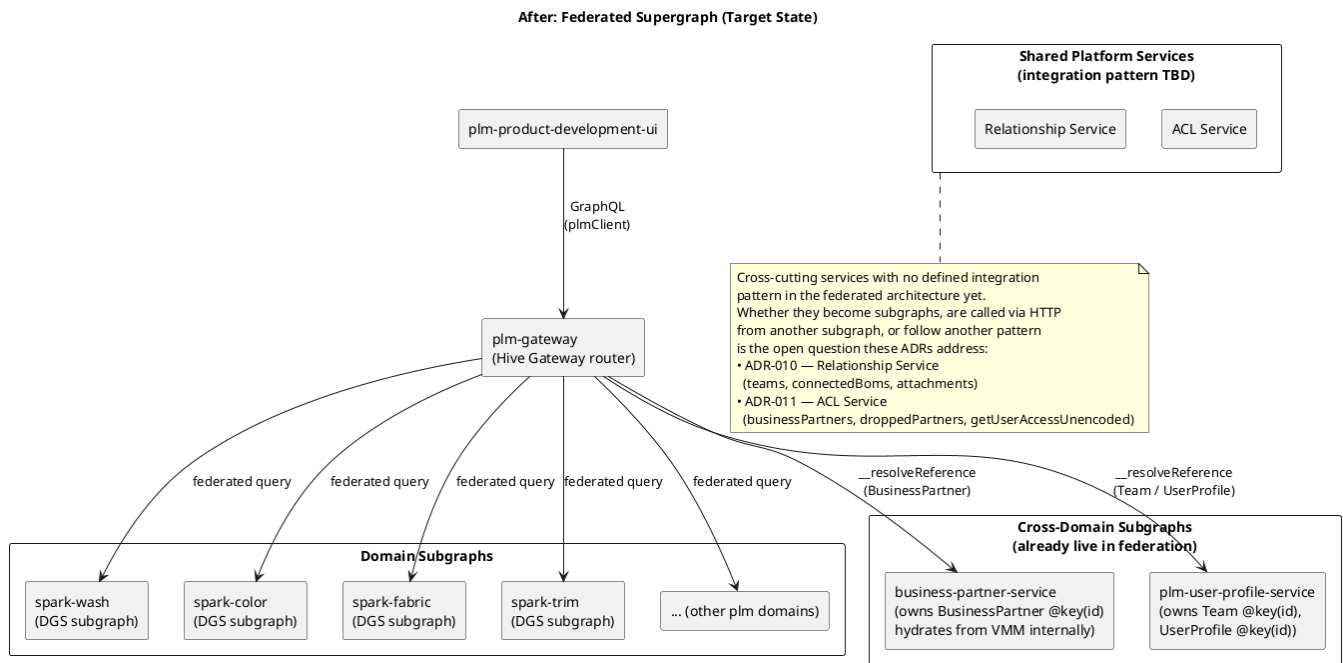
Before: Legacy Monolith

The legacy platform (`spark-internal-graphql`) is a single Node.js GraphQL server. All resolution logic — including ACL lookups, VMM hydration, and Relationship Service traversal — lives in shared utilities called directly from resolvers. The UI talks to one endpoint.



After: Federated Supergraph (Target State)

The target platform composes all 8 domain services as independent DGS subgraphs behind a **Hive Gateway** router. The UI migrates from `internalClient` to `plmClient`. Each subgraph owns its SDL and resolves its own fields — but the shared ACL/VMM/RelationshipService resolution logic from the monolith has **no federated home yet**. That is the open question both ADR-010 and ADR-011 address.



The Gap

Field	Monolith resolution	Federated home	ADR
businessPartners	loadBusinessPartners ACL VMM	Not yet decided	ADR-011
droppedPartners	loadDroppedBusinessPartners ACL	Not yet decided	ADR-011
teams	ACL Relationship Service	Not yet decided	ADR-010
connectedBoms	Relationship Service traversal	Not yet decided	ADR-010
getUserAccessUnencoded	ACL Service direct call	Not yet decided	ADR-011
BusinessPartner entity	VMM hydration	business-partner-service	—
Team entity	User profile service	plm-user-profile-service	—